

Background Information

In the past decade, the use of biodegradable plastics has dramatically risen. However, there is limited research on their harmfulness compared to non-biodegradable plastics. Both biodegradable and non- biodegradable plastics can create microplastics found on land and in water. These microplastics created are a danger to ocean and freshwater ecosystems affecting marine and freshwater organisms. The macroinvertebrate *Daphnia magna* is a commonly used model organism in ecotoxicology studies. Heart rate, egg count, and size are common stress indicators when experimenting with *Daphnia magna*.

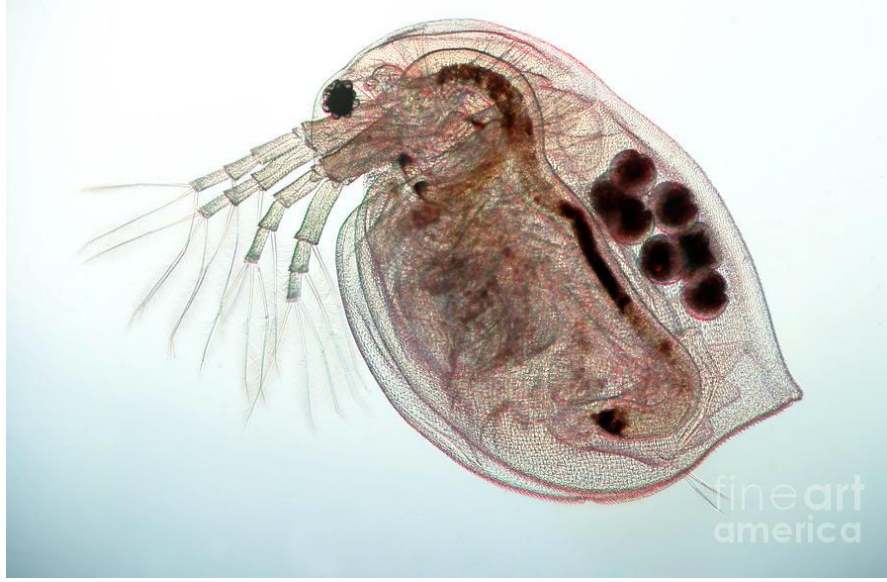
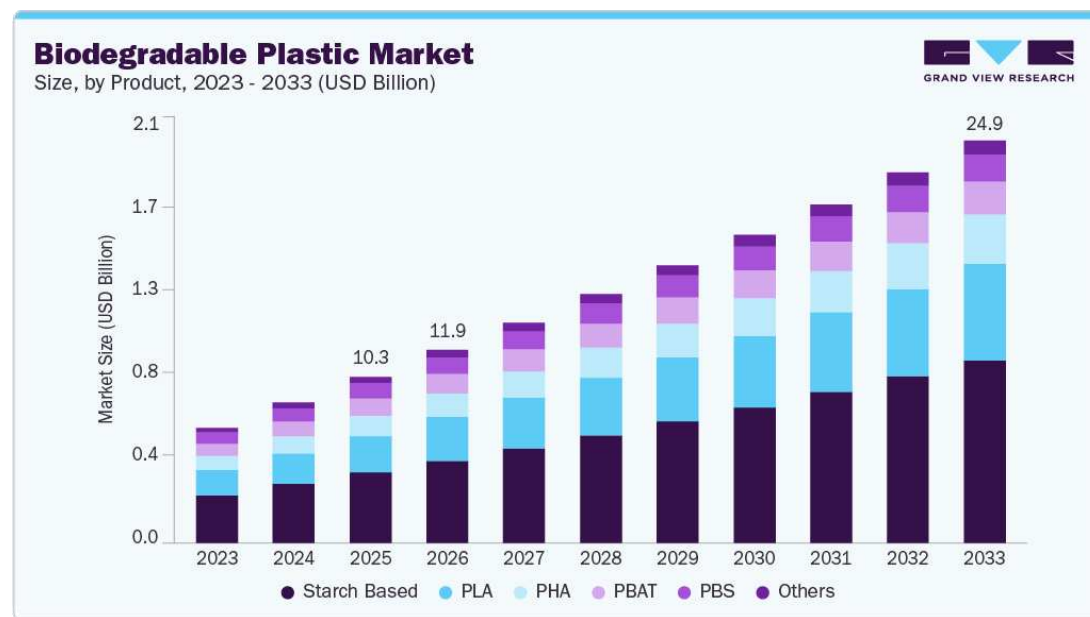


Figure 1: Image of female *Daphnia magna* keystone species in freshwater ecosystems. Commonly used model organism in ecotoxicology studies.

Note: From Water Flea *Daphnia magna* AB Poster by Fine Art America, 2026. Source [https://prints.science.com/featured/8-water-flea-daphnia-magna-tes-kinsman.html?product=poster]. Copyright 2026, Fine Art America.

Figure 2: Histogram showing past, current, and predicted growth for biodegradable plastic market.



Note: From Biodegradable Plastic Market by Grand View Research, 2026, source [https://www.grandviewresearch.com/industry-analysis/biodegradable-plastics-market]. Copyright 2026, Grand View Research.

Purpose, Hypothesis, and Variables

Purpose: This study aims to compare the biodegradable microplastic PLA and the non-biodegradable microplastic ABS on the freshwater macroinvertebrate *Daphnia magna* to discover how biodegradable microplastics differ towards non-biodegradable microplastics.

Hypothesis: The microplastics made from ABS non-biodegradable plastic will have a greater negative effect on the *Daphnia magna* than the microplastics made from the PLA biodegradable plastic because ABS contains toxic chemical compounds such as styrene and PLA is plant based.

Independent Variable: The type of microplastic the *Daphnia magna* are exposed to.

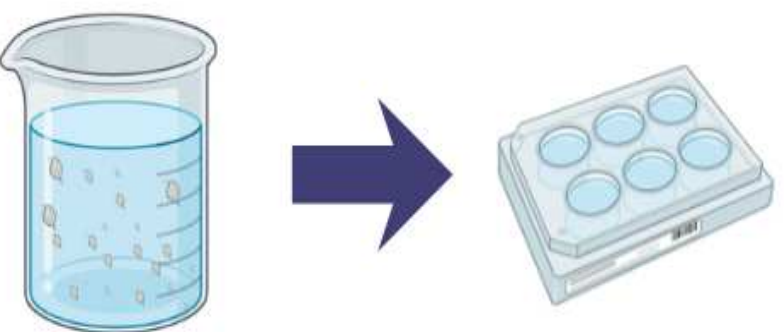
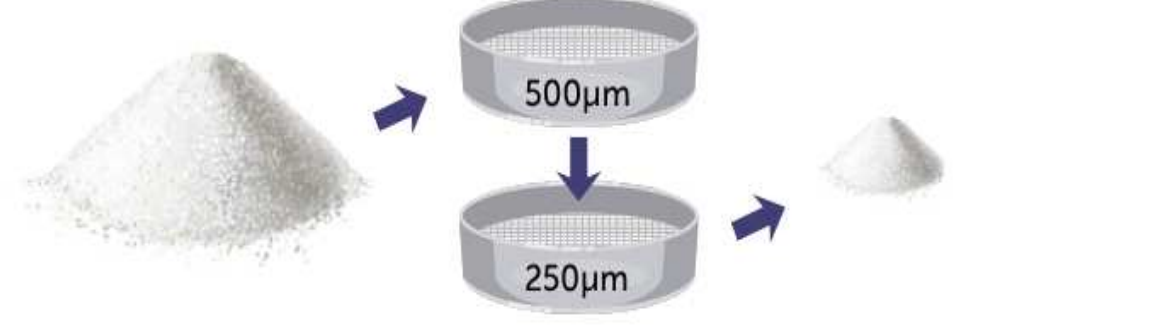
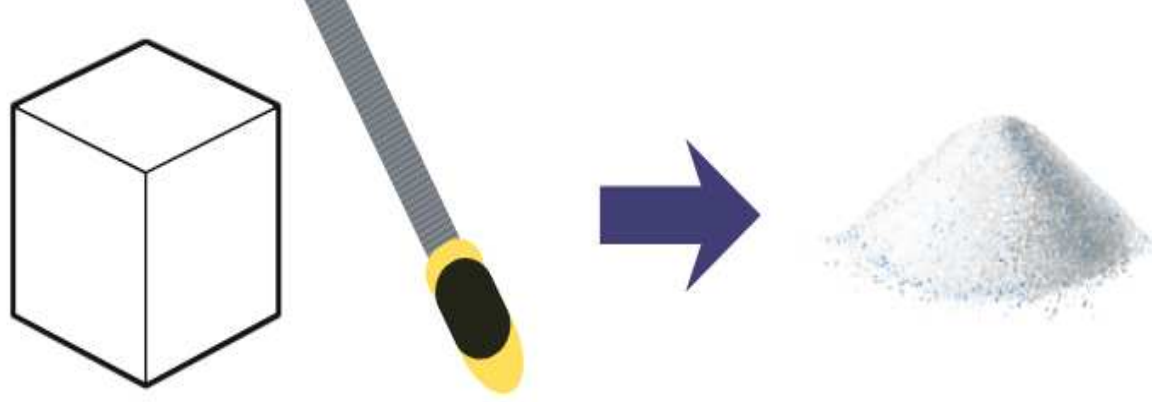
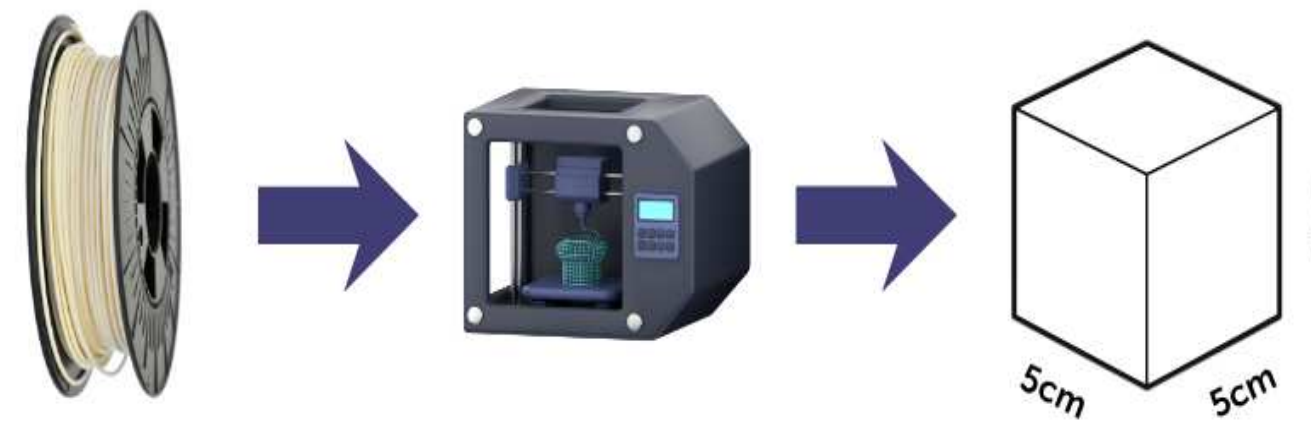
Dependent Variable: The effect it has on the heart rate, size, and egg counts of the *Daphnia magna*.

Controlled Variables: *Daphnia* are raised in spring water and feed on green algae. They are kept under a heat lamp while not attended to.

Comparing the Effects of the Biodegradable Microplastics PLA and ABS on the Heart Rate, Size, and Reproduction of *Daphnia magna*

Materials and Methods

- 3D Printer Filament is 3D printed into a 5x5x5 cm cube
- 3D Printed cube is filed with a metal filer into a coarse powder
- The powdered plastic is then filtered through two sieves to get microplastics between the sizes of 250-500µm.
- Microplastics are added into a beaker containing 20 daphnia and 1L of spring water containing algae at a concentration of 50mg/L.
- After 5 days of exposure to the microplastics, 9 daphnia will be taken out and put into a well plate.
- The well plate is then examined under a light and dissecting microscope and recorded with a phone camera in order to collect data.

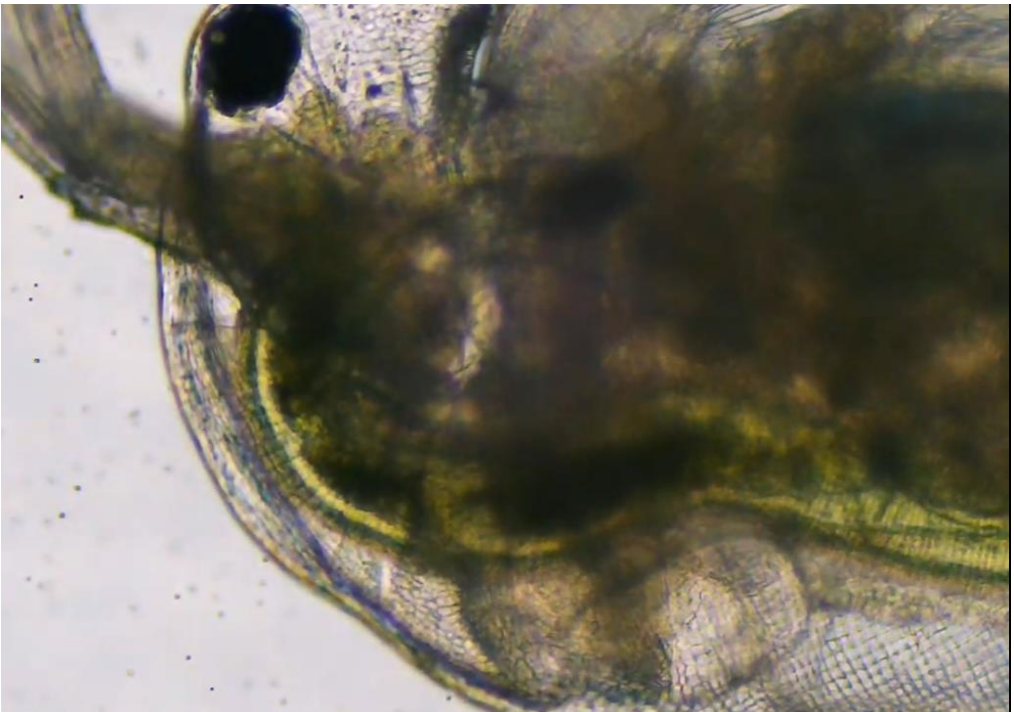


Note: Graphic made with BioRender and Canva

Figure 3: The materials used in the experiment, printer filament, filters, 3D printed filament, and brush. Microplastics made are shown in bags.



Figure 4: *Daphnia magna* under the light microscope.



Results

Figure 5: Scatter plot diagram that shows the relationship between the heart rate of *Daphnia magna* and the ABS, PLA, and no microplastic groups. Graph includes a box and whisker plot to compare data distribution.

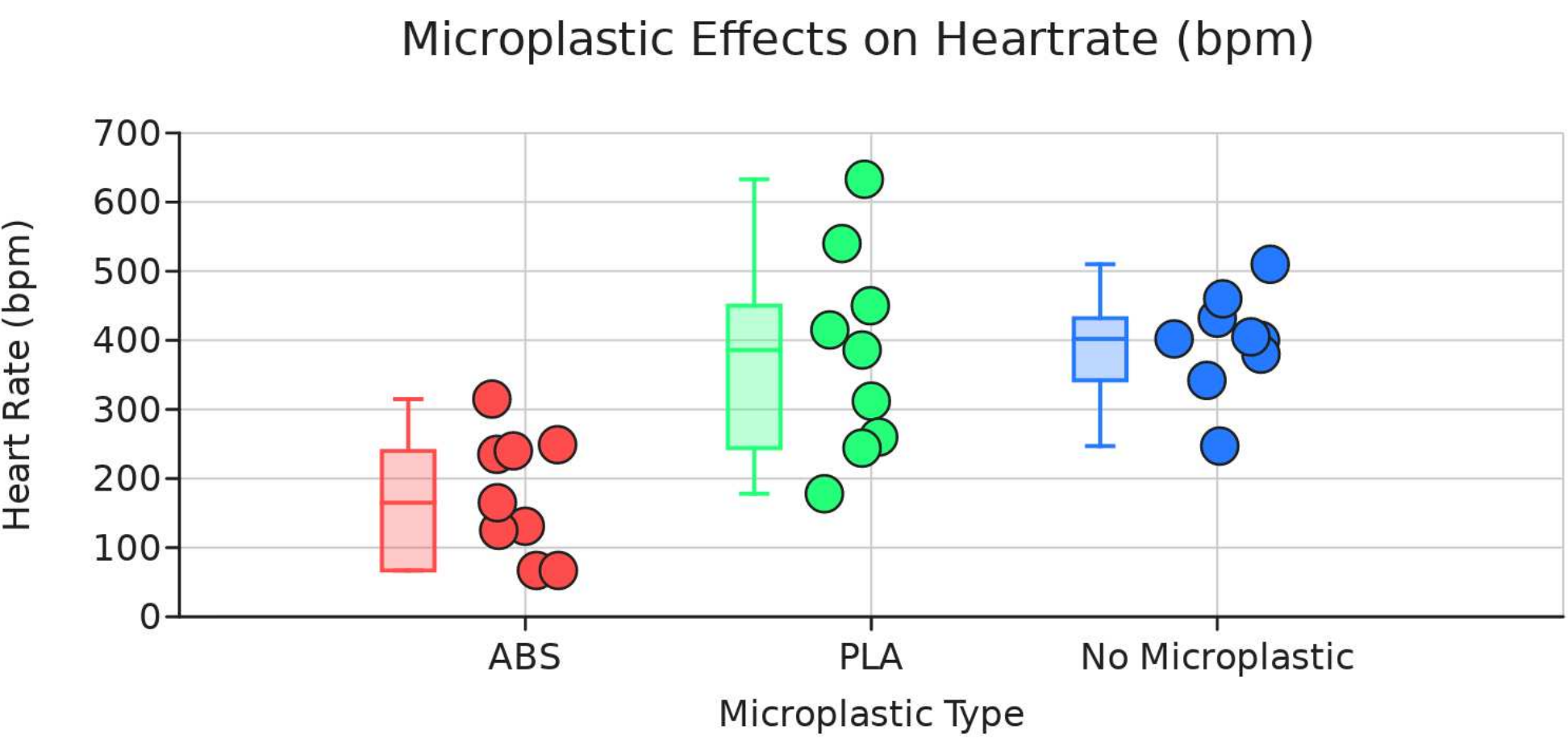


Figure 6: Scatter plot diagram that shows the relationship between the sizes of *Daphnia magna* and the ABS, PLA, and no microplastic groups. Graph includes a box and whisker plot to compare data distribution.

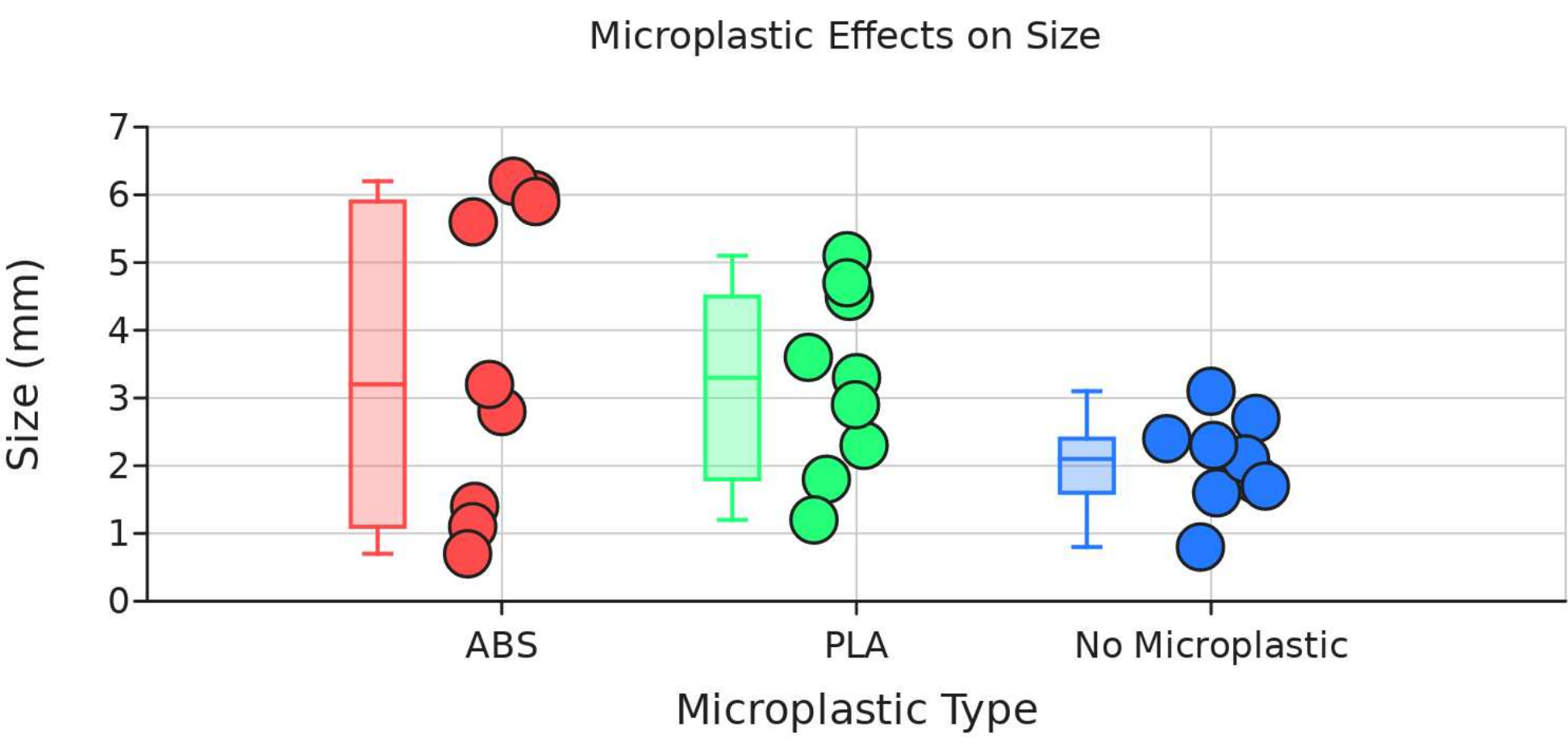
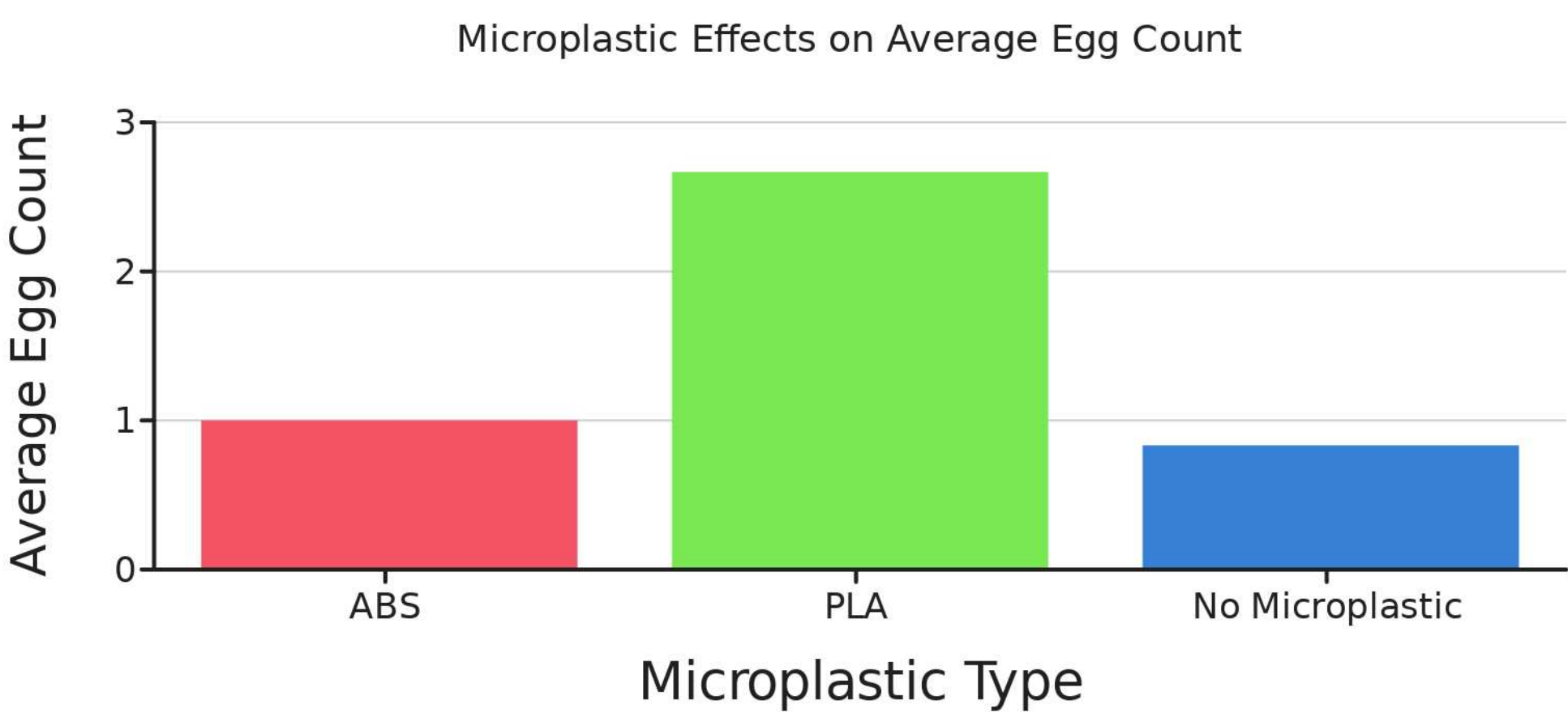


Figure 7: Histogram that shows the average amount of eggs per female *Daphnia magna* for each of the ABS, PLA, and no microplastic groups.



Note: Graphical representation made using DataClassroom

Conclusion

Statistical Tests:

To analyze the data collected during this experiment, a non-parametric Mann Whitney U test was conducted to compare the effects the two independent variables had on the heart rate, size, and average egg count in the *Daphnia magna*. The results found that there was a statistically significant difference between the effects which microplastics made from PLA and ABS had on the heart rate of *Daphnia magna* ($p < 0.01$). Additionally, the study found that microplastics made from PLA and ABS did not have statistically significant differences in their effect towards the number of eggs produced ($p = 0.79$) and the size of *Daphnia magna* ($p = 0.92$).

Conclusion

Exposing *Daphnia magna* to microplastics caused negative effects on heart rate. PLA microplastics showed mixed effects with higher heart rate than ABS and a large range of heart rates showing how it both raised and lowered heart rate. ABS microplastics showed greater impact on heart rate compared to PLA microplastics with a far lower average heart rate. Both microplastics showed similar effects on the size of the *Daphnia magna*, however ABS had a greater range in size. *Daphnia magna* exposed to PLA microplastics had an overall higher egg count. The control group showed microplastics had a greater negative effect on *Daphnia magna* supporting to conclusion that microplastics are an environmental pollutant. The results of this study contributes to the limited research done comparing the use of biodegradable microplastics to non-biodegradable microplastics on living organisms and paves the way for additional research to be done on this topic.

References

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- Graphs made using DataClassroom